Climate-Adaptive Smart Crop Monitoring and Automatic Irrigation System using STM32

1. Abstract

This project presents a climate-adaptive crop monitoring system using the STM32 Nucleo-F401RE microcontroller. The system monitors temperature, humidity, and soil moisture in real time using an HTU21D sensor and a soil moisture sensor. Timer-based sampling is used for environmental monitoring, while interrupt-driven logic is used for irrigation control. Based on soil conditions, the system automatically controls a pump through a relay and generates disease-risk alerts using a buzzer. Sensor readings and system status are continuously displayed through UART, providing a simple and efficient smart farming solution.

2. Introduction

Modern agriculture requires efficient water management and continuous monitoring of environmental conditions. Traditional irrigation methods often lead to water wastage and delayed response to changing climate conditions. This project aims to automate irrigation and provide environmental monitoring using embedded systems technology.

3. Objectives

- Monitor temperature and humidity using HTU21D sensor.
- Detect soil moisture conditions in real time.
- Automatically control irrigation based on soil moisture.
- Generate disease-risk alerts under unfavorable environmental conditions.
- Display system status through UART communication.
- Implement interrupt-driven embedded system design.

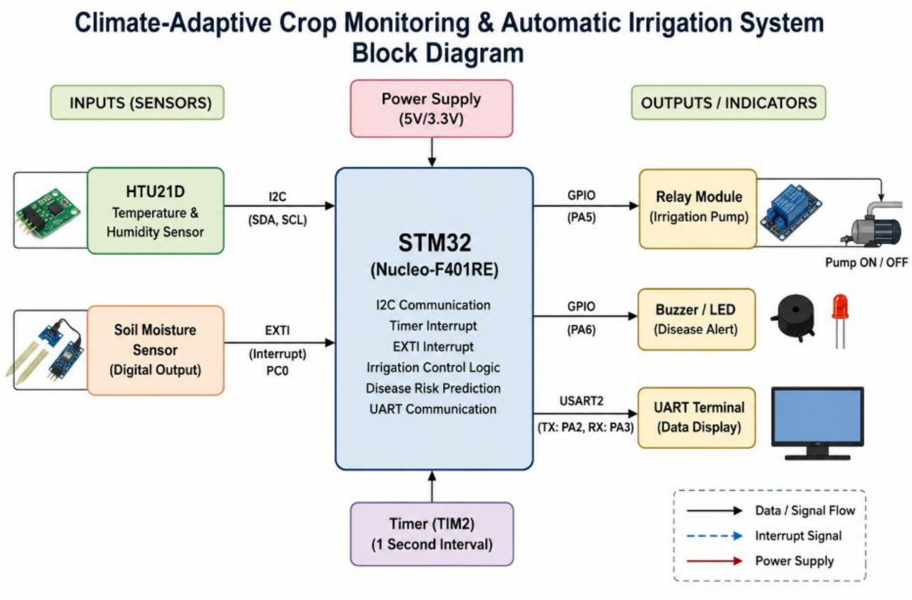
4. Hardware Components

- STM32 Nucleo-F401RE
- HTU21D Temperature and Humidity Sensor
- Soil Moisture Sensor
- Relay Module
- Buzzer
- USB-UART Interface
- Connecting Wires and Breadboard

5. Software Tools

- STM32CubeIDE
- STM32CubeMX
- Tera Term / PuTTY
- Embedded C
- STM32 HAL Library

6. System Architecture



7. Working Principle

1. TIM2 generates an interrupt every second.
2. STM32 reads temperature and humidity from HTU21D through I2C.
3. Soil moisture sensor generates an EXTI interrupt when moisture status changes.
4. If soil becomes dry, relay activates irrigation.
5. If soil becomes wet, relay turns irrigation OFF.
6. Disease risk is predicted based on temperature and humidity thresholds.
7. Sensor readings and system status are transmitted through UART.

9. Results

- Successful I2C communication with HTU21D sensor.

- Real-time soil moisture detection using EXTI interrupts.
- Automatic relay control for irrigation.
- Disease-risk alert generation using buzzer.
- Continuous UART monitoring.

10. Conclusion

The proposed climate-adaptive crop monitoring system successfully automates irrigation and environmental monitoring using STM32. The project demonstrates the practical use of I2C communication, timer interrupts, external interrupts, UART communication, and embedded automation in smart agriculture applications.

11. Future Scope

- MQTT-based cloud monitoring
- Mobile application integration
- Wi-Fi connectivity using ESP8266
- Data logging and analytics
- Machine learning-based disease prediction
- Weather forecast integration